

Robustness and Performance of Adaptive Suppression of Unknown Periodic Disturbances

Saeid Jafari, Petros Ioannou, Ben Fitzpatrick, and Yun Wang

Abstract—In recent years, a class of adaptive schemes has been developed for suppressing periodic disturbance signals with unknown frequencies, phases, and amplitudes. The stability and robustness of these schemes with respect to inevitable unmodeled dynamics and noise disturbances in the absence of persistently exciting signals has not been established despite successful simulation results and implementations. The purpose of this technical note is to propose a robust adaptive scheme for rejection of unknown periodic components of the disturbance and analyze its stability and performance properties. First, we consider the ideal case (non-adaptive) when complete information about the characteristics of the disturbance is available. We show that the rejection of periodic terms may lead to amplification of output noise and in some cases lead to a worse output performance. The way to avoid such undesirable noise amplification is to increase the size of the feedback control filter in order to have the flexibility to achieve rejection of the periodic disturbance terms while minimizing the effect of the noise on the output. The increased filter order leads to an over-parameterized scheme where persistence of excitation is no longer possible, and this shortcoming makes the use of robust adaptation essential. With this important insight in mind, the coefficients of the feedback filter whose size is over parameterized are adapted using a robust adaptive law. We show analytically that the proposed robust adaptive control scheme guarantees stability, performance and robustness with respect to unmodeled dynamics and bounded broadband noise disturbances. We use numerical simulations to demonstrate the results.

Index Terms—Over-parameterization, robust adaptive control, unknown periodic disturbances.

I. INTRODUCTION

Adaptive control techniques have already been used to suppress the effect of disturbances in many different applications that include acoustic noise cancellation [1] and mechanical vibration control [2], [3] as well as laser beam control problems such as wavefront compensation in adaptive optics [4]–[6] and active pointing and tracking [7], [8].

These applications motivated a series of theoretical results for band limited disturbance rejection presented in a wide range of references such as [9]–[17] based on internal model, frequency weighted methods, variable order least squares lattice filters and other techniques.

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S. Jafari and P. Ioannou are with the Department of Electrical Engineering, University of Southern California, Los Angeles, CA 90089 USA.

B. Fitzpatrick is with the Department of Mathematics, Loyola Marymount University, Los Angeles, CA 90045 USA, and also with Tempest Technologies, Los Angeles, CA 90045 USA.

Y. Wang is with Tempest Technologies, Los Angeles, CA 90045 USA.

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While the aforementioned results provide efficient techniques to suppress the effect of unknown disturbances, which are demonstrated in simulations and experiments, adaptive techniques such as these have been known to fail [16] in some cases. In almost all past work in the area, the problem of robustness with respect to plant unmodeled dynamics and bounded noise disturbances has not been adequately addressed. In fact, most adaptive laws proposed in literature for the problem of disturbance rejection were shown not to be robust back in the early 80's [18]. These issues indicate the importance of establishing the robustness of these schemes with respect to inevitable modeling errors.

The purpose of this technical note is to propose a robust adaptive control scheme for disturbance rejection and show analytically its stability and performance in the presence of unmodeled dynamics and noise. For the sake of clarity, the presentation of the technical note is for SISO discrete-time systems; the results however are extendable for MIMO case. The technical note is organized as follows. Section II introduces the notation and definitions used throughout the technical note. In Section III, the problem is defined and formulated. Section IV presents the main results of the technical note. Finally, concluding remarks are summarized in Section V.

II. PRELIMINARIES AND NOTATIONS

An m -input, q -output finite-dimensional LTI system with transfer function $H(z)$, input $u(k)$, and output $y(k)$ may be expressed as $y(k) = H(z)[u(k)]$. For a signal x , the truncated function x_k is defined as $x_k(i) = x(i)$, for $0 \leq i \leq k$ and $x_k(i) = 0$, for $i > k$. The $\mathcal{H}_\infty$ and $\mathcal{L}_1$ norms of a system with transfer matrix $H(z)$ are defined as $\|H(z)\|_\infty = \max_{\Omega \in [0, \pi]} \sigma_{\max}(H(e^{j\Omega}))$, where $\sigma_{\max}$ denotes the maximum singular value, and $\|H(z)\|_1 = \max_{i=1}^q \sum_{j=1}^m \|h_{ij}\|_1$, where h_{ij} is the impulse response corresponding to the ij -element of $H(z)$, and $\|h_{ij}\|_1 = \sum_{k=0}^{\infty} |h_{ij}(k)|$. We say signal x is μ -small in the mean square sense and write $x \in \mathcal{S}(\mu)$, when $(1/N) \sum_{i=k}^{k+N-1} x(i)^T x(i) \leq c_0 \mu + c_1/N$, $\forall k, N$, and some constant $c_0, c_1 \geq 0$ that are independent of the constant μ [18], [19].

III. CONTROL SYSTEM MODEL

We consider stable plants of the form

$$y(k) = G(z)[u(k)] + d(k) \\ = G_0(z)(1 + \Delta_m(z))[u(k)] + d(k) \quad (1)$$

where $z = e^{j\omega T_s}$ is the operator of the z -transformation, T_s is the sampling period, $y(k) = y(kT_s)$ is the observed output, $u(k)$ is the control input, $d(k)$ is an unknown (and unobserved) bounded disturbance. The plant discrete-time model $G(z)$ (a representation of the sampled continuous-time model combined with the zero-order hold) is typically not perfectly known. We model $G_0(z) = Z_0(z)/R_0(z)$, where $R_0(z)$ is a monic Hurwitz polynomial of degree n , as a known estimate of $G(z)$ (possibly non-minimum phase) with $\Delta_m(z)$ representing an unknown multiplicative uncertainty such that $\Delta_m(z)G_0(z)$ is proper with stable poles. A wide class of plant modeling errors can

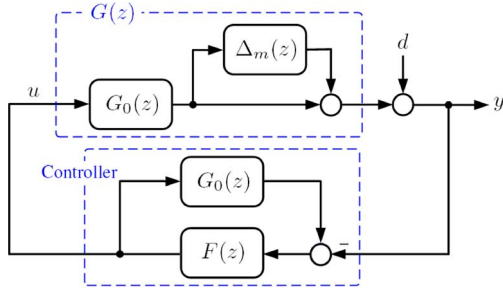


Fig. 1. Structure of the discrete-time closed-loop system.

be expressed in the form of (1) [18]. In many, if not most, control applications, the plant has already been stabilized with a fixed-gain controller, and the plant (1) is an augmentation of the baseline original control design (see [7], [11], and [12]). We assume that the disturbance term $d(k)$ is modeled as

$$d(k) = \sum_{i=1}^{n_\omega} \kappa_i \sin(\omega_i T_s k + \varphi_i) + v(k) = d_s(k) + v(k) \quad (2)$$

where the parameters κ_i , ω_i , φ_i , and the number of distinct frequencies n_ω are all unknown and v is a bounded random noise disturbance with amplitude much smaller than that of d_s . It should be noted that the selection of sampling period T_s depends on the characteristics of the disturbance such as the upper bound on the highest significant frequency present in d_s in (2).

The control objective is to minimize the effect of d on the plant output y and make the norm of y as small as possible. Fig. 1 illustrates the structure of the closed-loop system in which the stable strictly proper filter $F(z)$ must be designed to achieve the objective.

From Fig. 1 and the small-gain theorem [20], a sufficient condition for stability of the closed-loop system is that

$$\|\Delta_m(z)G_0(z)F(z)\| < 1 \quad (3)$$

where $z = e^{j\Omega}$, $\Omega = \omega T_s \in [0, \pi]$, and $\|\cdot\|$ can be any norm satisfying the multiplicative property such as $\mathcal{H}_\infty$ norm or $\mathcal{L}_1$ norm [20]. It should be noted that in general condition (3) may not be necessary for stability of the closed-loop system; it is a sufficient condition and states that nonzero but small size modeling errors can be tolerated in the feedback control system. Clearly, with $F(z) = G_0^{-1}(z)$ for any modeling error satisfying $\|\Delta_m\| < 1$, the plant output will be zero for any disturbance with any characteristics; however, such a filter is not realizable. In the next section, we present the design and analysis of a realizable finite impulse response (FIR) filter structure for disturbance suppression in the presence of unmodeled plant dynamics and bounded noise.

IV. CONTROLLER DESIGN

In this section, we begin with the ideal case where the frequencies of the disturbance $d_s(k)$ in (2) are assumed to be known exactly. The known frequency case allows us to identify what is the best performance that can be achieved and set the reference performance and robustness levels to be compared with in the more realistic case where the frequencies are unknown and changing with time.

A. Known Disturbance Case

This subsection considers the case when the frequencies of the disturbance are known. The analysis of the closed-loop system in this case provides insights that can be used to deal with the unknown disturbance case using robust adaptive control techniques.

Let F be an N th-order FIR filter of the form $F(z, \theta) = \Theta(z)/z^N$, where $\Theta(z) = \theta_0 + \theta_1 z + \dots + \theta_{N-1} z^{N-1}$, which can be written as

$$F(z, \theta) = \sum_{i=0}^{N-1} \theta_i z^{i-N} = \theta^T \alpha(z) \quad (4)$$

where $\theta = [\theta_0, \theta_1, \dots, \theta_{N-1}]^T \in \mathbb{R}^N$ and $\alpha(z) = [z^{-N}, z^{1-N}, \dots, z^{-1}]^T$. The control objective is to find the parameter vector θ so that the norm of y is minimized.

If the frequencies of d_s are known, then its internal model $D_s(z) = \prod_{i=1}^{n_\omega} (z^2 - 2\cos(\omega_i T_s)z + 1)$ is also known. It is obvious that if (3) is satisfied and the zeros of the sensitivity transfer function H from $d = d_s + v$ to y given by

$$y = H(z)[d] \triangleq \frac{1 - G_0 F}{1 + \Delta_m G_0 F} [d_s + v] \quad (5)$$

contain the roots of D_s , then the effect of d_s on y will be completely eliminated.

Lemma 1: Consider the filter (4) and the transfer function H in (5). For any $G_0 = Z_0/R_0$, there exist a vector θ so that the zeros of H contain the roots of D_s , if and only if Z_0 and D_s are coprime and $N \geq n_d$, where $n_d = \deg(D_s)$. Furthermore, the vector θ is unique if $N = n_d$.

Proof: The proof is given in the Appendix.

If $N = n_d$, the unique θ satisfying the conditions of Lemma 1 may lead to a large peak magnitude of $G_0 F$. This situation is not desirable, because it may lead to noise amplification, and the performance may be much worse than the case with no control input. Moreover, from (3), the size of the modeling error which can be tolerated may be too small leading to smaller stability margins.

In order to completely reject the effect of the periodic components of the disturbance on y , improve the robustness with respect to unmodeled dynamics without amplifying the output noise, we choose a higher order filter, $N > n_d$. In this case, we have an infinite number of vectors θ that achieve the rejection of the disturbance periodic terms and therefore have more flexibility in selecting the best parameter vector that fits all the criteria, i.e. rejecting the periodic terms of the disturbance without amplifying the noise effect and reducing the impact on the robust stability margins. At this point, we may look for vectors θ from this set that minimize the objective function

$$J(\theta) = \|G_0 F\|_\infty = \left\| \frac{Z_0 \Theta}{z^N R_0} \right\|_\infty, \text{ s.t. } Z_0 \Theta + D_s L = z^N R_0 \quad (6)$$

for some polynomial $L(z)$. In other words, we find a θ such that at $z_i = e^{j\omega_i T_s}$, $F(z_i, \theta) = G_0^{-1}(z_i)$ and $J(\theta)$ is minimized. The existence of a minimizer is the subject of the following lemma.

Lemma 2: Consider the closed-loop system shown in Fig. 1. Assume the filter is of the form (4) and the output disturbance can be represented as in (2). If Z_0 and D_s have no common factors and $N > n_d$, then there exists a θ with $|\theta| \leq r$, for some $r > 0$, for which the effect of d_s on y is completely rejected and the cost function (6) is minimized.

Proof: The proof is given in the Appendix.

Lemma 2 shows the existence of a constant parameter vector θ for complete rejection of the periodic components of disturbance signals of the class defined by (2). It also provides a smaller maximum peak of the magnitude of $G_0 F$ which is equivalent to minimizing the effect of noise on the output. The following theorem gives the properties of the closed-loop system.

Theorem 1: Consider the closed-loop system shown in Fig. 1 with the filter (4). Assume the frequencies of d_s are known, $N \geq n_d$, and θ^* is a solution of the optimization problem (6). If the inequality

$$\|\Delta_m(z)G_0(z)F(z, \theta^*)\|_\infty < 1 \quad (7)$$

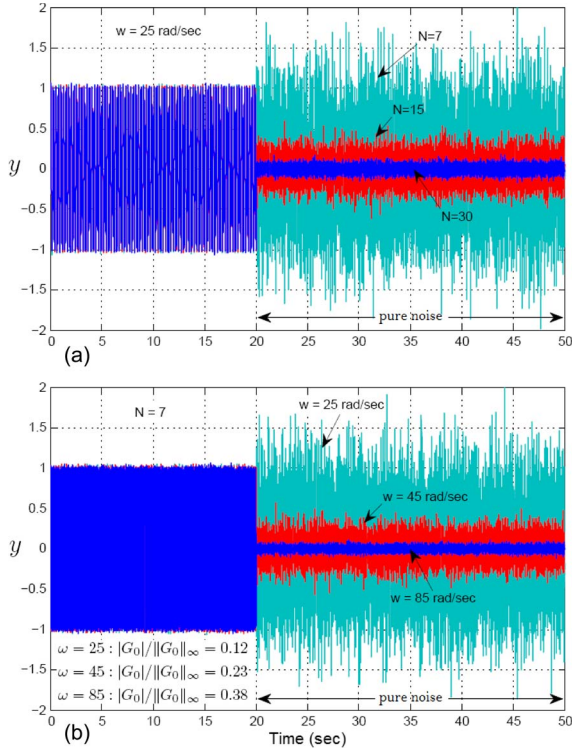


Fig. 2. Control performance for the case of known disturbance frequencies. The control input is applied at $t = 20$ sec. (a) Fixed disturbance frequency $\omega = 25$ rad/sec and different filter orders N ; increasing N improves the performance. (b) Fixed filter order $N = 7$ and different disturbance frequencies.

holds, then

$$\lim_{T \rightarrow \infty} \sup_{k \geq T} |y(k)| \leq \left\| \frac{1 - G_0(z)F(z, \theta^*)}{1 + \Delta_m(z)G_0(z)F(z, \theta^*)} \right\|_1 v_0 \leq c(1 + \|G_0(z)F(z, \theta^*)\|_1) v_0 \quad (8)$$

where $v_0 = \sup_k |v(k)|$ and c is a finite positive constant. In addition, in the absence of the noise (i.e., when $v_0 = 0$), the plant output $y(k)$ goes to zero exponentially fast with time.

Proof: The proof is given in the Appendix.

Theorem 1 shows that we can design a filter which completely rejects the effect of d_s on y . However, in the presence of noise, a large magnitude of the sensitivity function $H(z, \theta^*)$ given by (5) may lead to noise amplification, especially at high frequencies. The situation may be worse if the plant has very small gain at the frequency range of the disturbance and larger gain at high frequencies. In such case, the design of a pre-compensator to shape the frequency response of the plant to achieve a good compromise between performance and robustness will be a possible remedy. We illustrate these results using the following example.

Example 1: Consider the following 3rd-order model:

$$G_0(z) = \frac{-0.00146(z - 0.1438)(z - 1)}{(z - 0.7096)(z^2 - 0.04369z + 0.01392)} \quad (9)$$

Fig. 2 shows the performance of the off-line designed controllers obtained from (6) for complete rejection of the periodic term of a disturbance, where d_s is a single-frequency sinusoidal signal with amplitude one, v is a zero-mean Gaussian noise with standard deviation 0.02 and $|v| \leq 0.1$. In all simulations, the sampling period is $T_s = 1/480$ sec. In Fig. 2(a), the frequency of d_s is $\omega = 25$ rad/sec. The performance of the controllers with different orders N are shown with different colors.

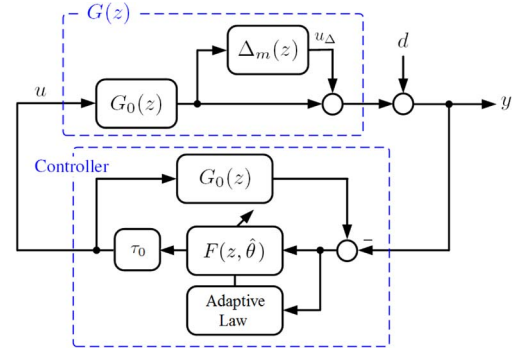


Fig. 3. Structure of adaptive control system.

After closing the loop at $t = 20$ sec, the sinusoidal part is completely rejected but the noise part is amplified for low order filters. As the order of the filter increases it provides the flexibility to reject the sinusoidal disturbance terms without amplifying the noise. Fig. 2(b) shows the performance for different disturbance frequencies $\omega = 25, 45$, and 85 rad/sec, where the filter order $N = 7$ is fixed. The plant has a zero at $z = 1$ and the low plant gain (and low coprimeness of $Z_0(z)$ and $D_s(z)$) at the low frequency of 25 rad/sec makes it difficult to reject the disturbance. In fact, the performance is worse than that of the open-loop system due to the amplification of the output noise. For higher frequencies however the gain of the plant corresponding to these frequencies is higher leading to a better performance as shown in Fig. 2(b).

The control design and analysis of the known frequency case is very useful as it establishes analytically that for better performance the order of the filter F has to be much larger than n_d in order to provide the structural flexibility to design the coefficients of the filter to simultaneously reject the periodic components of the disturbance and minimize the effect of the output noise on the plant output. This analysis provides the form and dimension of the controller filter F in the adaptive case where the frequencies of the disturbances are completely unknown. We treat this case in the following subsection.

B. Unknown Disturbance Case

In practice, the characteristics of the disturbance signal are unknown and may be time-varying, so an adaptive control scheme is required to meet the control objective. In this subsection, our aim is to design and analyze an adaptive FIR filter of the form

$$F(z, \hat{\theta}(k)) = \sum_{i=0}^{N-1} \hat{\theta}_i(k) z^{i-N} = \hat{\theta}^T(k) \alpha(z) \quad (10)$$

in the feedback loop, modifying Fig. 1 to include adaptation, as shown in Fig. 3, where τ_0 is a nonzero scalar. The adaptively tuned filter (10) is designed to minimize the output variance, suppressing to the extent possible the disturbance d . From Fig. 3, we have $u = -\tau_0 F(z, \hat{\theta})[\zeta]$, where $\zeta = y - G_0(z)[u]$, and $\hat{\theta}$ is the estimate of the unknown optimum parameter vector θ^* which at time $t = kT_s$ is $\hat{\theta}(k-1)$. Then, we write the control signal as $u(k) = -\tau_0 \hat{\theta}^T(k-1)w(k)$, where $w = \alpha(z)[\zeta]$. The plant output can be written as $y = G_0(z)[u] + \zeta = -\tau_0 G_0(z)[\hat{\theta}^T w] + \zeta$. From the Swapping Lemma [18] we have $G_0(z)[\hat{\theta}^T w]_k = \hat{\theta}^T G_0(z)[w]_k + G_{0c}(z)[(G_{0b}(z)[w^T]) \Delta \hat{\theta}]_k$, where $\Delta \hat{\theta}(k) = \hat{\theta}(k+1) - \hat{\theta}(k)$, $G_{0c}(z) = -C_0^T(zI - A_0)^{-1}$, $G_{0b}(z) = z(zI - A_0)^{-1}B_0$, and the quadruple (A_0, B_0, C_0, D_0) represents a minimal realization of G_0 . Then, we rewrite y as $y = -\hat{\theta}^T \tau_0 G_0(z)[w] - \eta_0 + \zeta = -\hat{\theta}^T \tau_0 G_0(z)[w] - \eta_0 + \eta$, where $\eta_0 = \tau_0 G_{0c}(z)[(G_{0b}(z)[w^T]) \Delta \hat{\theta}]$,

where $\eta = (1 - \tau_0 G_0(z)F(z, \theta^*))[v + u_\Delta]$, $u_\Delta = \Delta_m(z)G_0(z)[u]$, and $\hat{\theta} = \hat{\theta} - \theta^*$, where θ^* is such that $1 - \tau_0 G_0(z)F(z, \theta^*)$ has zero gain at the frequencies of the disturbance. The existence of such a desired vector θ^* has already been established in the previous section. From the above presentation, we have the following parametric model:

$$\zeta(k) = \theta^{*T} \phi(k) + \eta(k) \quad (11)$$

where $\zeta = y - G_0(z)[u]$ and $\phi = \tau_0 G_0(z)[\alpha(z)[\zeta]]$ are available for measurement and the unknown parameter θ^* appears linearly with respect to the known regressor signal vector ϕ . The error term η is driven by the modeling error and noise and may also depend on exponentially decaying to zero terms due to initial conditions that for the sake of clarity are considered to be zero as they do not affect the analysis.

From the design and analysis of the known frequency case we require $N > n_d$ in order to have the structural flexibility for the adaptive control law to choose the parameters that reject the periodic disturbance terms and minimize the effect of the noise. Since the number of frequencies of d_s is unknown, a conservative upper bound for this number is used. This choice leads to an over-parametrization, and the regressor vector ϕ in (11) cannot be persistently exciting (PE) even when the periodic disturbance terms provide adequate excitation [18]. For this reason the adaptive law used to update the estimate of the parameter vector θ^* needs to be robust, otherwise the presence of noise and modeling error terms may cause parameter drift as pointed out in [18]. This is an interesting case as even though PE can be guaranteed by the design due to the richness of the disturbance, which in turn can guarantee PE of a minimal dimensional regressor, our analysis of the known frequency case dictates that we have to over-parameterize and therefore reject the powerful property of PE in order to achieve better disturbance rejection properties.

The parametric model (11) can be used to develop a wide class of robust parameter estimators using the approach in [19] as follows. If $\hat{\theta}(k-1)$ is the estimate of θ^* at time $k-1$, the predicted value of the signal $\zeta(k)$ can be generated as $\hat{\zeta}(k) = \hat{\theta}^T(k-1)\phi(k)$. We define a normalized estimation error as $\varepsilon(k) = (\zeta(k) - \hat{\zeta}(k))/m^2(k)$, where $m^2(k) = 1 + c_0 \phi^T(k)\phi(k)$, with $c_0 > 0$, is a normalizing signal. To develop an adaptation scheme for $\hat{\theta}$, we consider the robust pure least-squares algorithm [19]

$$P(k) = P(k-1) - \frac{P(k-1)\phi(k)\phi^T(k)P(k-1)}{m^2(k) + \phi^T(k)P(k-1)\phi(k)}$$

$$\hat{\theta}(k) = \text{proj}(\hat{\theta}(k-1) + P(k)\varepsilon(k)\phi(k)) \quad (12)$$

where $P(0) = P^T(0) > 0$ and projection operator $\text{proj}(\cdot)$ is used to guarantee that $\hat{\theta}(k) \in \mathbb{S}$, $\forall k$, where $\mathbb{S}$ is a compact set defined as $\mathbb{S}(\theta_{\max}) = \{\mathbf{x} \in \mathbb{R}^N \mid \mathbf{x}^T \mathbf{x} \leq \theta_{\max}^2\}$, where $\theta_{\max} > 0$ is chosen so that the coefficient vector of the optimum filter, θ^* , belongs to $\mathbb{S}$. Since we do not have *a priori* knowledge on the norm of θ^* , $\theta_{\max}$ must be chosen sufficiently large. The projection may be implemented as

$$\bar{\theta}(k) = \hat{\theta}(k-1) + P(k)\varepsilon(k)\phi(k)$$

$$\bar{\rho}(k) = P^{-1/2}(k)\bar{\theta}(k)$$

$$\rho(k) = \begin{cases} \bar{\rho}(k) & \text{if } \bar{\rho}(k) \in \bar{\mathbb{S}} \\ \bar{\rho}'(k) = \perp \text{ proj of } \bar{\rho}(k) \text{ on } \bar{\mathbb{S}} & \text{if } \bar{\rho}(k) \notin \bar{\mathbb{S}} \end{cases}$$

$$\hat{\theta}(k) = P^{1/2}(k)\rho(k) \quad (13)$$

where P^{-1} is decomposed into $(P^{-1/2})^T(P^{-1/2})$ at each instant k to guarantee all properties of the corresponding recursive least-squares

algorithm without projection. The set $\mathbb{S}(\theta_{\max})$ is transformed to $\bar{\mathbb{S}}$ such that if $\bar{\theta}(k) \in \mathbb{S}(\theta_{\max})$, then $\bar{\rho}(k) \in \bar{\mathbb{S}}$. Since $\mathbb{S}$ is chosen to be convex and $\bar{\rho}(k) = P^{-1/2}(k)\bar{\theta}(k)$ is a linear transformation, then $\bar{\mathbb{S}}$ is also a convex set [21].

An alternative to projection is a fixed σ -modification where no bounds for $|\theta^*|$ are required [18]. An additional constraint that is often used in practice to ensure that the control signal u is within certain limits $\pm u_{\max}$ is a saturation block inserted right after $F(z, \hat{\theta})$ in Fig. 3. The above adaptive control law in contrast to those proposed in literature for similar applications has all the robustness modifications to guarantee stability in the presence of modeling errors something that needs to be analytically established as we will show below.

In implementing (12), we also need to use covariance resetting modification [19] and monitor the covariance matrix $P(k)$ to make sure it does not become small in some directions, i.e. its minimum eigenvalue is always greater than a small positive constant. Such modifications are presented and analyzed in [18], [19] and other references on adaptive control.

The performance of this adaptive system can be guaranteed as is detailed in the following theorem in which it is assumed that the characteristics of the disturbance are unknown, but there exists a θ^* (unknown to designer) to achieve the best performance.

Theorem 2: Consider the adaptive control law (11) and (12), and choose $N > 2\bar{n}_\omega$, where $\bar{n}_\omega$ is an upper bound for the number of distinct frequencies in the range of interest in the sinusoidal part of the disturbance. If the modeling error satisfies $\bar{\theta}_M \|\tau_0 \Delta_m(z)G_0(z)\|_1 < 1$, where $\bar{\theta}_M = \max_{\theta \in \mathbb{S}} \|\theta\|_1$, then all signals in the closed-loop system are uniformly bounded and the plant output satisfies

$$\frac{1}{T} \sum_{i=k}^{k+T-1} |y(i)|^2 \leq c(\mu_\Delta^2 + v_0^2) + \frac{c}{T}, \quad \forall k \geq 0, \forall T > 1 \quad (14)$$

and some finite constant c independent of T, k , where μ_Δ is a constant proportional to the size of the unmodeled dynamics, and $v_0 = \sup_k |v(k)|$. In addition, in the absence of modeling error and noise (i.e., when $\Delta_m = 0$ and $v_0 = 0$), the adaptive control law guarantees the convergence of $y(k)$ to zero.

Proof: The proof is given in the Appendix.

Theorem 2 states that the average energy of the plant output is of the order of the size of the modeling error and the noise level. It follows from (14) that as T goes to infinity the term c/T that is due to initial conditions also goes to zero. In the absence of the modeling error and noise, the output converges to zero despite the presence of completely unknown periodic components. We demonstrate the performance of the proposed robust adaptive control scheme using the following example where we assume unknown frequencies, amplitudes, and phases of the periodic terms in the disturbance.

Remark 1: The use of positive scalar τ_0 is to increase the level of the regressor signal vector ϕ which is beneficial to convergence. A large value of τ_0 , however, reduces the stability margin given by Theorem 2. This trade off can be improved by replacing the constant τ_0 , with a strictly proper stable filter $\tau_0(z)$ which can be designed to improve convergence and have a positive rather than negative impact on robustness. In addition, $\tau_0(z)$ can be chosen to improve the gain of $G_0(z)$ when the zeros of $G_0(z)$ are very close to the zeros of the generating polynomial of the disturbance. These details are currently under investigation and will be presented in a future paper.

Example 2: Consider the plant model (9). For simulation purposes, we consider the multiplicative uncertainty $\Delta_m(z) = -0.0002/(z + 0.99)^2$ whose magnitude (for $z = e^{j\omega T_s}$) is of the order of -80 dB for $\omega < 900$ rad/sec and increases to 6 dB as $\omega \rightarrow \pi/T_s = 1508$ rad/sec. The sampling period is $T_s = 1/480$ sec, $N = 50$,

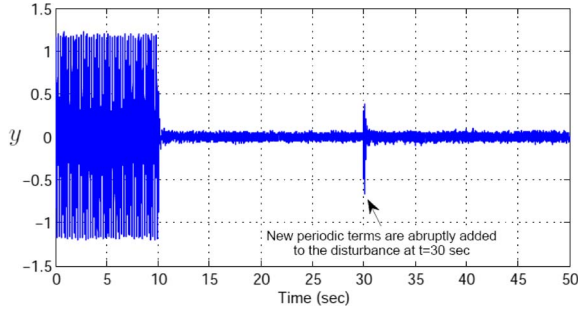


Fig. 4. Performance of the adaptive scheme. The control input is applied at $t = 10$ sec. At $t = 30$ sec new sinusoidal terms are abruptly added to the disturbance; the adaptive controller updates its parameters at this time to suppress the effect of new terms.

$c_0 = 1$, $P(0) = 20I$, $\tau_0 = 100$, v is a zero-mean Gaussian noise with standard deviation 0.02, and $|v| \leq 0.1$. The periodic term of the disturbance that we like to reject is $d_s(t) = 0.7 \sin(25t + (\pi/3)) + 0.5 \sin(225t + (\pi/4))$. Fig. 4 shows the performance of the adaptive scheme, where at $t = 10$ sec the feedback is switched on and at $t = 30$ sec new sinusoidal terms $0.6 \sin(85t - (\pi/6))$ and $0.4 \sin(125t + (\pi/2))$ are added to the disturbance. It is clear from simulations that the feedback control rejects the periodic terms in the plant output and adjusts itself to reject new ones at $t = 30$ sec automatically with a very small transient despite the presence of noise and unmodeled dynamics.

V. CONCLUSION

In this technical note, we propose a robust adaptive scheme for attenuation of disturbances dominated by periodic terms and analyze its stability and performance properties in the presence of unmodeled dynamics and noise disturbances. First, we consider the ideal case (non-adaptive) when complete information about the characteristics of the disturbance is available. This step demonstrates the level of performance that can be achieved in the case of known disturbances and establishes the form and order of the controller filter that provides disturbance rejection as well as noise effect attenuation. Then, we investigate the adaptive case when the periodic components of the disturbance are unknown by using the same form and order for the controller filter as in the known frequency case and updating its parameters using a robust adaptive law. Our results show under what conditions the unknown periodic terms can be suppressed and what parameters contribute to the performance of the overall adaptive control scheme. In addition, we show analytically that the proposed robust adaptive control scheme guarantees stability, performance, and robustness in the absence of persistently exciting signals.

APPENDIX

Proof of Lemma 1: To prove this lemma, we need to establish that the existence of polynomials $\Theta(z)$ and $L(z)$ satisfying $Z_0\Theta + D_sL = z^N R_0$ is equivalent to Z_0 and D_s being coprime with $N \geq n_d$. The proof follows by writing the above polynomial equation as a Sylvester-type matrix equation $S\mathbf{x} = \beta$, where $\mathbf{x} = [\theta^T, l^T]^T$ and matrix S contains the coefficients of Z_0 and D_s , vector β contains the coefficients of $z^N R_0$, and vectors θ, l are unknown coefficients of Θ and L , respectively (see [22, Ch. 2]). $\square$

Proof of Lemma 2: The existence of a θ for which we can achieve complete rejection of the sinusoidal part of the disturbance is guaranteed from Lemma 1. We define $Q = \{\theta : \exists l \mid S\mathbf{x} = \beta\}$,

where $S, \mathbf{x}$, and β are defined in the proof of Lemma 1. The set Q is nonempty by Lemma 1 and affine by inspection, and hence closed and convex. The function J is continuous and convex. As such, it attains its minimum on any compact set. The problem of minimizing (6) can be written as an LMI optimization problem and can be solved by using the Matlab YALMIP Toolbox. $\square$

Proof of Theorem 1: The existence of a minimizer θ^* for the constrained optimization problem (6) is guaranteed by Lemma 1, 2. Then, from (5), and using the small gain theorem [20], if for all frequencies $\|G_0(z)F(z, \theta^*)\Delta_m(z)\|_\infty < 1$, then the stability of the closed-loop system is guaranteed. Since with θ^* the zeros of $1 - G_0F$ contain the roots of D_s , then $y(k) = H(z)[v(k)] + \epsilon_0(k)$, where $\epsilon_0(k)$ is an exponentially decaying term. Thus, $\lim_{T \rightarrow \infty} \sup_{k \geq T} |y(k)| \leq \|H(z)\|_1 v_0$. The second upper bound follows by noting that $\|H\|_1 \leq \|1/(1 + \Delta_m G_0 F)\|_1 \|1 - G_0 F\|_1 \leq c(1 + \|G_0 F\|_1)$, where c is a generic symbol for a positive constant. This completes the proof. $\square$

Proof of Theorem 2: From Fig. 3, the signal ζ can be written as $\zeta = y - G_0(z)[u] = u_\Delta + d = -\Delta_m(z)G_0(z)[\tau_0 \hat{\theta}^T w] + d$, then $\|\zeta_k\|_\infty \leq \|\tau_0 \Delta_m(z)G_0(z)\|_1 \|(\hat{\theta}^T w)_k\|_\infty + d_0$, where $d_0 = \sup_k |d(k)|$. The projection operator ensures the boundedness of $\hat{\theta}$ and we can write $\|(\hat{\theta}^T w)_k\|_\infty \leq \max_{i=0}^k \|\hat{\theta}(i-1)\|_1 \|w_k\|_\infty \leq (\max_{\hat{\theta} \in \mathbb{S}} \|\hat{\theta}\|_1) \|w_k\|_\infty \leq \bar{\theta}_M \|\alpha(z)\|_1 \|\zeta_k\|_\infty = \bar{\theta}_M \|\zeta_k\|_\infty$. Thus, $\|\zeta_k\|_\infty \leq \bar{\theta}_M \|\tau_0 \Delta_m(z)G_0(z)\|_1 \|\zeta_k\|_\infty + d_0$. Then, if the modeling uncertainty term is such that $\bar{\theta}_M \|\tau_0 \Delta_m(z)G_0(z)\|_1 < 1$, then $\zeta(k)$ and therefore all signals in the closed-loop system are uniformly bounded.

The normalized estimation error can be expressed as $\varepsilon(k) = (-\phi^T(k)\hat{\theta}(k-1) + \eta(k))/m^2(k) = (y(k) + \eta_0(k))/m^2(k)$. From (12), $P(k)$ is bounded and the covariance resetting ensures that $P(k)$ is positive definite. It can be shown (see [19, Ch. 4]) that the parameter error $\bar{\theta}$ can be written as $\bar{\theta}(k) = P(k)P^{-1}(k-1)\bar{\theta}(k-1) + P(k)\phi(k)\eta(k)/m^2(k)$. Consider the Lyapunov-like function $V(k) = (\bar{\theta}(k) - \theta^*)^T P^{-1}(k)(\bar{\theta}(k) - \theta^*) = (\bar{\rho}(k) - \rho^*)^T (\bar{\rho}(k) - \rho^*)$ whose boundedness is guaranteed due to the projection and covariance resetting, where $\rho^* = P^{-1/2}(k)\theta^* \in \mathbb{S}$.

If $\bar{\rho}(k) \in \mathbb{S}$, $V(k) = \bar{\theta}^T(k)P^{-1}(k)\bar{\theta}(k)$, then we have $\Delta V(k) = V(k) - V(k-1) = -v_0(k)\varepsilon^2(k)m^2(k) + v_1(k)\varepsilon(k)m(k)(\eta(k)/m(k)) + v_2(k)(\eta^2(k)/m^2(k))$, where $v_0(k) = 1/(1 + \bar{\phi}^T(k)P(k-1)\bar{\phi}(k))$, $v_1(k) = 2(v_0(k) + \bar{\phi}^T(k)P(k)\bar{\phi}(k) - 1)$, $v_2(k) = 2 - v_0(k) - \bar{\phi}^T(k)P(k)\bar{\phi}(k)$, and $\bar{\phi}(k) = \phi(k)/m(k)$. Then, from the boundedness of $\bar{\phi}(k)$ and $P(k)$, it follows that $\Delta V \leq -\kappa_0 \varepsilon^2 m^2 + \kappa_1 |\varepsilon m| |\eta/m| + 2|\eta/m|^2$, where $\kappa_0 = 1/(1 + \bar{\phi}_0 \lambda_{\max}(P_0))$, $\bar{\phi}_0 = \sup_k \|\bar{\phi}(k)\|^2$, $\kappa_1 = 2/\kappa_0$ are positive constants. Using the inequality $-ax^2 + bx \leq -(a/2)x^2 + b^2/(2a)$ we have $\Delta V \leq -(\kappa_0/2)\varepsilon^2 m^2 + (2 + \kappa_1^2/(2\kappa_0))\eta^2/m^2 \leq -c\varepsilon^2 m^2 + c\eta^2/m^2$, where c is a generic symbol for a positive constant. The error term in (11) is $\eta = (1 - \tau_0 G_0(z)F(z, \theta^*)) [v + u_\Delta]$, then $|\eta/m|^2 \leq c(v_0^2 + \mu_\Delta^2) \triangleq \bar{\eta}^2$, for some finite $c > 0$. Thus, $\Delta V(k) \leq -c\varepsilon^2(k)m^2(k) + c\bar{\eta}^2$ and by taking summation on both sides, we obtain $(1/T) \sum_{k=1}^T \varepsilon^2(k)m^2(k) \leq c\bar{\eta}^2 + (c/T)(V(0) - V(T)) \leq c\bar{\eta}^2 + c/T$, that is $\varepsilon m \in \mathcal{S}(\bar{\eta}^2)$. Since $m^2 \geq 1$, we have $\varepsilon^2 \leq \varepsilon^2 m^2$, therefore $\varepsilon \in \mathcal{S}(\bar{\eta}^2)$. From (12), $\Delta \hat{\theta} = P\varepsilon\phi = P\varepsilon m\bar{\phi}$, then $|\Delta \hat{\theta}|^2 \leq \lambda_{\max}^2(P) |\varepsilon m|^2 |\bar{\phi}|^2 \leq \lambda_{\max}^2(P_0) \bar{\phi}_0 |\varepsilon m|^2$ which implies that $|\Delta \hat{\theta}| \in \mathcal{S}(\bar{\eta}^2)$.

If $\bar{\rho}(k) \notin \mathbb{S}$, then we have $V(k) = (\bar{\rho}'(k) - \rho^*)^T (\bar{\rho}'(k) - \rho^*)$, where $\bar{\rho}'(k)$ is an orthogonal projection of $\bar{\rho}(k)$ on $\mathbb{S}$. Then, since $\rho^* \in \mathbb{S}$, $(\bar{\rho}'(k) - \rho^*)^T (\bar{\rho}'(k) - \rho^*) \leq (\bar{\rho}(k) - \rho^*)^T (\bar{\rho}(k) - \rho^*) = \bar{\theta}^T(k)P^{-1}(k)\bar{\theta}(k)$. This enables us to establish all properties of the adaptive law without projection [21].

Since w is bounded and $\Delta \hat{\theta} \in \mathcal{S}(\bar{\eta}^2)$, then $\eta_0 \in \mathcal{S}(\bar{\eta}^2)$. Also, from $y = \varepsilon m^2 - \eta_0$, we have $|y|^2 \leq 2\bar{m}^2 |\varepsilon m|^2 + 2|\eta_0|^2$, where $\bar{m}^2$ is an upper bound for m^2 , which implies $y \in \mathcal{S}(\bar{\eta}^2)$. This completes the proof of Theorem 2. $\square$

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